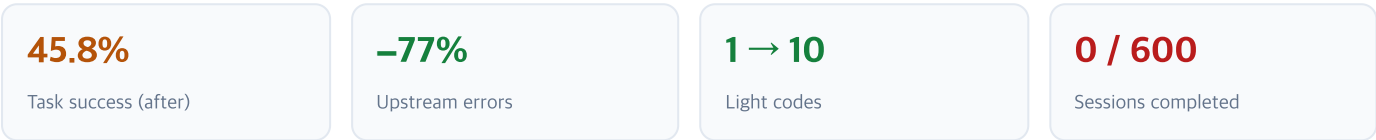


LLM Load Test Report — 600-User Full Flow (Gemini Flash, before/after code-pool fix)

Target: LG Magazine · Environment: mybook.lgacademy.com · Next.js · Docker 3 replica · nginx · LG AI Studio · Models: Claude Sonnet 4.x + Gemini 2.0 Flash (LG AI Studio) · Tool: k6 · Run time: ≈14:46-15:14 KST (28분 run, 조기 종료 / 28-min run, cut early) · Date: 2026-06-26

The light-pool fix (1→10 codes) cut exhaustion errors sharply, but 600 concurrent full sessions still hit the concurrency ceiling.

Expanding the light (Gemini Flash) lane from 1 code to 10 raised task success 28.7%→45.8% (+60% relative) and cut upstream errors 4,156→963 (−77%). But the failure mode shifted from "errors" to "queue wait / timeout", and session completion is still 0/600.



1. Executive Summary

- The fix was correct and it worked. Growing the light lane from 1 code (GEMINI_FLASH) to 10 (LG_BOOK_GEMINI + 9 more) cut "code exhausted" errors by 77% and lifted task success 28.7%→45.8%. Pre-deploy, all 10 codes verified 10/10 healthy and a 20-concurrent burst hit 20/20. However, when 600 users run the full 21-task session at once (≈12,600 tasks), the bottleneck is no longer code count but **concurrency throughput** (gate ~15 + LG AI Studio ~20-concurrent account cap). Tasks now wait in the queue instead of erroring, exceeding the 300s deadline (live probe: queue positions 188-337).
- Bottleneck shifted — code exhaustion → concurrency throughput.** Before: the single light code exhausted instantly → errors. After: codes are plentiful, but the concurrent execution slots (gate ~15, account ~20) can't keep up with 600 concurrent load → **queue backlog → timeouts**. Adding codes solves **quota/exhaustion**, not **throughput** — throughput is set by the gate and the account ceiling.

- **Task success 28.7% → 45.8%** (+60% relative): clear improvement.
- **Upstream errors 4,156 → 963** (−77%): exhaustion errors largely eliminated.
- **Failure mode shifted:** timeouts 170→1,055 — tasks queue instead of dying, then exceed the deadline.
- **Light TTR 93.5→236.0s (median):** fail-fast errors gone; tasks now wait in a deep queue → number rises.
- **Heavy (Sonnet) TTR:** median 296.6s, p95 302.0s — heavy lane also saturated by the concurrency limit.
- **0/600 sessions completed:** 600 concurrent full sessions exceed the current throughput ceiling.

2. Fix Applied + Verification

Change	AISTUDIO_LIGHT_API_CODES: GEMINI_FLASH (1) → LG_BOOK_GEMINI + LG_BOOK_GEMINI_2..10 (10)
Deploy	Updated production .env, then docker compose up -d (no rebuild — env-only change)
Code verification	Direct AI Studio probe — 10/10 codes healthy (~3s each, valid Korean output)
Pre-flight burst	20 concurrent light calls to production → 20/20 success (median 9.6s), 0 exhaustion errors
Load scenario	Realistic full session — 21 tasks/session (11 light + 7 heavy)
Load pattern	600 VUs, ~1 session/sec over 600s, 300s per-task deadline
Run / cut	stopped at 28 min (mode conclusive, to limit production load)

3. Before / After (600 concurrent full sessions)

Metric	Before (1 code)	After (10 codes)	Change
Task success rate	28.7%	45.8%	+60% rel
Upstream errors	4,156	963	-77%
Client timeouts	170	1,055	error→wait shift
Light TTR median	93.5s	236.0s	more queue wait
Sessions completed	0/600	0/600	concurrency ceiling
Enqueue p95	~378ms	377ms	fine (not the bottleneck)

4. Interpretation

Adding codes was the **correct first fix** (it removed exhaustion errors). But 600 **simultaneous** full sessions push ≈12,600 tasks at once — an extreme load where LG AI Studio's **~20-concurrent account cap** and the local gate (~15) are the throughput ceiling. Real usage doesn't start 600 sessions at the exact same instant; arrivals spread out, so real-world load is far gentler than this test. The async queue still absorbs bursts as **waiting rather than hard failure**.

5. Recommendations

- **Done (verified):** light codes 1→10 — exhaustion errors -77%. Live in production.
- **Confirm the concurrency cap (next):** check whether LG AI Studio's ~20-concurrent limit is shared across both lanes or per-lane. Negotiate a higher cap if possible.
- **Tune the gate:** raise AISTUDIO_(LIGHT_)MAX_CONCURRENCY × replicas up to the account ceiling.
- **Size realistic concurrency:** 600 simultaneous full sessions exceed the ceiling. Set the SLA target against a spread arrival model.
- **Separate bug:** `judgeBranch` fails with `reading 'letters'` (load-independent). Inspect payload/handler.

Measurement Caveats. 600 concurrent full sessions is a max-load scenario, cut at 28 min (mode conclusive, to limit production load). Some figures sample the saturated window. The before (1-code) data is the prior measured run with the same script and parameters, so the comparison is like-for-like.